

Milestone Accomplishment Record

River Hawk Consulting, LLC · IMAX HCD Prototype OT
August 8, 2026

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Milestone Accomplishment Record

Agreement No. 5600002690012 · River Hawk Consulting, LLC · UNCLASSIFIED Period of performance: **6 – 20 August 2026** (Article II)

Submitted under **Article VIII(b)**: *"Attachment 2 provides the mutually agreed-to acceptance criteria for each milestone. The Performer shall document for acceptance by the Government, the accomplishments of each milestone and shall submit such documentation to the AOR and AO for approval."*

Attachment 2 was not included in the executed agreement. The body references it four times (Article VIII(b) twice, Article XII once) but the LIST OF ATTACHMENTS names only Attachment 1. In its absence, the milestones and success indicators scored below are taken from the **TDD's own Milestones / Success Indicators table**, which is Attachment 1 and therefore incorporated. River Hawk has requested Attachment 2 from the AO and will re-score against it on receipt if it differs. See `signed_ot_audit.md` §2.3.

Calendar

TDD Assumption (1): *"Day 1 begins the first calendar day after agreement execution."* The agreement was executed **6 August 2026**, so Day 1 is **7 August** and Day 14 is **20 August**, consistent with Article II.

Milestone	Day	Date	Status
M1 — proto-personas, empathy maps, task flows	2	Sat 8 Aug	✅ complete
M2 — repo scaffolded; five mock services containerized and smoke-deployed	4	Mon 10 Aug	🟡 see below
M3 — full wireframe set validated and frozen as build baseline	7	Thu 13 Aug	✅ complete
M4 — feature-complete UI wired to mock services; smoke suite passing	11	Mon 17 Aug	✅ complete
M5 — all six data deliverables transmitted; live demonstration provided	14	Thu 20 Aug	🕒 scheduled

River Hawk developed the demonstration software **at its own expense before award** (TDD, IP Assertion). The period of performance therefore opened with the build already complete, and M1, M3 and M4 were satisfied by work predating their nominal dates. This is stated plainly rather than presented as early delivery: the Government is receiving the benefit of pre-award investment, which is what the TDD said it would.

M1 — Proto-personas & empathy maps · Day 2

Success indicator (TDD): *"Two proto-personas with empathy maps and task flows delivered for Sponsor review."* **Verification (TDD)**: *"Artifacts in repo (optional Touchpoint 1 offered ~Day 3)."*

Evidence	Where
Four personas — two required, two secondary	<code>hcd/personas.md</code>
Empathy maps (Says / Thinks / Does / Feels)	<code>hcd/empathy_maps.md</code>
Four task flows and the end-to-end journey map	<code>hcd/journey_map.md</code>
20 per-persona success criteria, each naming the specification that replays it	<code>hcd/journey_map.md</code>

Delivered as `project/delivery/hcd_artifacts.pdf`. Exceeds the indicator: two personas were required, four are delivered.

Touchpoint 1 was offered and remains offered; the Sponsor has not scheduled it.

M2 — Repo, containerized services, sandbox smoke deployment · Day 4

Success indicator (TDD): *"GitHub repo scaffolded; all five mock services containerized and smoke-deployed to Platforma (or matched local K8s if sandbox access is pending)."* **Verification (TDD)**: *"Deployment log; running pods; repo access."*

The TDD wrote this verification assuming direct cluster access. The target environment is driven by **ArgoCD**, and River Hawk has no `kubectl` against it — deployment is a declarative sync, not a command someone runs. The equivalent evidence, and what should be captured instead, is:

TDD wording	ArgoCD equivalent
Deployment log	The Application's sync history — revision, sync result, timestamp
Running pods	The Application's health status and resource tree, showing all six workloads <code>Healthy</code>
Repo access	The Application's source repository and target revision

A single `argocd app get imax -o yaml` captures all three, or a screenshot of the Application view if only the UI is reachable. This is the same evidence in the form the environment actually produces it, and the Helm chart at `deploy/chart/` is what ArgoCD consumes.

[m2_evidence_checklist.md](#) is the working copy for capturing it: the command, what a good result looks like, what each likely failure means, and what to do with a partial result.

Evidence	Where
GitHub repository, CI on every push	<code>Blackfell-Group/imax-chat-viewer</code>
Five schema-pinned mock services	<code>routes/</code> , <code>deploy/k8s/mock-*.yaml</code>
Dockerfiles and Kubernetes manifests	<code>deploy/Dockerfile.{spa, mock}</code> , <code>deploy/k8s/</code>
Helm chart for ArgoCD, proven equivalent to the manifests	<code>deploy/chart/</code> , <code>scripts/compare-chart-kustomize.py</code>
Six-image set built, scanned (trivy CRITICAL gate), deployed to a cluster and smoked — every push	<code>.github/workflows/angular-ci.yml</code>
Pod sizing within the TDD's 0.5 vCPU / 512 MB ceiling	250m CPU / 256Mi memory on all six

Status 🟡. The TDD's stated fallback — *"matched local K8s if sandbox access is pending"* — is satisfied continuously and mechanically: CI re-proves the full chain on every push. Platforma sandbox access was not granted during the period.

A Platforma build has since been reported by River Hawk's own team. It is *not* scored here yet, because none of the equivalent evidence above has been captured. This row is completed by attaching an `argocd app get` output or the Application view; until then the documented fallback is what is claimed.

M3 — Wireframes validated and frozen · Day 7

Success indicator (TDD): *"Full wireframe set validated with Sponsor input and frozen as build baseline."* **Verification (TDD):** *"Baselined wireframes; internal approval record (optional Touchpoint 2)."*

Evidence	Where
Persona-indexed annotated wireframes, every prototype screen	<code>hcd/wireframes.md</code>
Pain-to-design traceability: 24 pains → response → screen → specification → screenshot	<code>hcd/traceability_matrix.md</code>
Assumption log, 16 entries, 6 closed	<code>hcd/assumption_log.md</code>
Three workflow-model findings with rationale and restoration paths	<code>hcd/linguist_workflow_model.md</code> , <code>hcd/bilingual_display_model.md</code> , <code>hcd/one_output_model.md</code>

Sponsor input was not available. Wireframes were validated against the documented assumption log — the fallback the TDD states for exactly this case — and every assumption is flagged for Sponsor review in the acceptance package. Touchpoint 2 remains offered.

M4 — Feature-complete UI, green smoke suite · Day 11

Success indicator (TDD): *"Feature-complete UI wired to mock services; smoke suite passing on the full mission loop."* **Verification (TDD):** *"Green Playwright run; build demo (optional Touchpoint 3)."*

Evidence	Where
Every wireframed screen implemented	<code>angular/src/app/features/</code>
Full mission loop: queue → search → thread → bilingual review → enrichment → OCR → promote to gold → export	<code>tests/ng/</code>
55 e2e specs green , plus 10 Angular unit and 59 Node unit specs	<code>npm run test:ng</code> , <code>npm test --prefix angular</code> , <code>npm run test:node</code>
Ported Phase 2 acceptance flows: 7 of 7 covered — 3 verbatim, 4 amended by logged HCD decisions	<code>project/evidence/README.md</code>
Machine-readable results	<code>project/evidence/e2e-results-*.json</code>

TDD Task 5 performance target verified. The TDD commits QA to *"<200 ms interaction latency on large logs via virtualization."*

`tests/ng/virtual-scroll.spec.js` synthesizes a 2,000-message thread, asserts the DOM holds a bounded window, and measures worst scroll-to-repaint at **~17 ms**. This was previously unverified and the virtualization itself was previously absent; both were found and closed by the 7 August audit.

Twelve defects from internal review were closed on 7 August, each with a specification: resizable panels; a resizable, maximizable and zoomable scan viewer; session-wide document tags with an explicit Add control; right-to-left scan rendering; chronological message ordering with an ASC/DESC toggle; copy confirmation with a working manual fallback; a flagged-for-targeter facet and sort; a larger resizable export dialog; drag-reorderable gold copy that the export follows; and a retractable review verdict. Logged as QA in the board export and burndown — **not** claimed as Sponsor validation.

M5 — Acceptance package and live demonstration · Day 14

Success indicator (TDD): *"All six data deliverables transmitted; live demonstration provided."* **Verification (TDD):** *"Acceptance package receipt."*

 **Scheduled for Thursday 20 August 2026, in person.** Transmittal and demonstration are both required; neither substitutes for the other.

Data deliverables — TDD D1–D6 crosswalk

The TDD names six data deliverables with formats and due days. This is the mapping the Government should use for acceptance.

Milestone Accomplishment Record

ID	Deliverable	Format required	Due	Artifact
D1	HCD Artifacts — personas, empathy maps, task flows/journey map, full wireframe set, assumption log, pain-to-design traceability matrix	PDF + source	D2 / D7 / D14	hcd/ (source) · project/delivery/hcd_artifacts.pdf
D2	UI Codebase — SPA implementing all HCD screens; Dockerfiles and Kubernetes charts for SPA and mock services; GitHub repository	Git repo	D4, then continuous	angular/, deploy/ on main; Blackfell-Group/imax-chat-viewer
D3	Test Suite — Playwright end-to-end smoke tests, test scripts, and final results report	In repo + PDF	D11 / D14	tests/ (source) · project/acceptance/test_results.md · project/delivery/test_results.pdf
D4	Documentation — user guide and developer guide, including production cutover path	PDF	D14	docs/ · project/delivery/{user_guide,developer_guide,deployment_guide,architecture}.pdf
D5	Project Artifacts — sprint-board export, burndown chart, risk register (final state)	PDF / CSV	D14	project/delivery/project_artifacts.pdf · project/delivery/board_export_final.csv
D6	Demo & Acceptance Package — recorded walkthrough (Sponsor receives unlimited rights),	MP4 / PDF + live	D14	project/acceptance/*.mp4 · project/delivery/acceptance_deck.pdf · live demo 20 Aug

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ID	Deliverable	Format required	Due	Artifact
	summary slide deck (objectives, test results, open issues), live demonstration			

Beyond D4. Article XII requires *"the architecture the Performer will deliver prior to the end of the period of performance"*, against the Article I(b) definition — sufficient *"to enable the Government to implement the system without requiring further development or support from the Performer."* Delivered as [docs/architecture.md](#) / [project/delivery/architecture.pdf](#). This is an Article XII obligation the TDD's D1–D6 list does not name; it was missed until the 7 August audit.

Administrative deliverables

Required by the agreement, outside the TDD's D1–D6.

Requirement	Article	Due	Status
FOCI package — SF 328, Key Management Personnel list, Organizational Entity Structure	XIV(c)	Not stated	 disposition requested from the AO
Cleared Personnel Certification Report (Attachment 3 template)	XV(n)	30 September annually	 disposition requested; expected nil return
Cleared Personnel Disposition Report (Attachment 3 template)	XV(o)	Within 60 days of completion (~19 Oct 2026)	 disposition requested; expected nil return

Article XV(a) scopes the security article to *"the extent that this Agreement involves access to national security information."* All work is unclassified and no personnel are cleared under this agreement, so these are expected to be nil returns — but a nil return is still a return, and the AO should confirm rather than River Hawk assume.